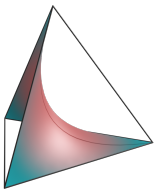
TUHH
Hamburg
University of
Technology



Project Work
for Mr. Alexander Kranz

B-3-2026
Student ID: 590156

Study Course: Data Science

Title: Low-dimensional Control under Embodiment Constraints: From LQR to Reinforcement Learning

Project description: We consider settings inspired by embodied intelligence, in which embodiment constraints limit the system's relevant dynamics to a low-dimensional structure, even though many sensor measurements and actuator inputs may be available. In such cases, it is natural to ask whether feedback controllers should also be low-dimensional or low-rank rather than fully parameterized. This project studies full-rank and low-rank feedback representations and compares their performance, robustness, and parameter efficiency in control and learning tasks. The study is motivated by recent work on structured policy optimization for LQR, data-enabled policy optimization, and the geometry of policy search in control [1, 2, 3], as well as by earlier ideas from embodied control and cheap control architectures [4, 5, 6].

The project is organized in two stages. First, the analysis is restricted to the linear-quadratic regulator (LQR) setting for linear state-space models, where fixed, learned, and model-informed low-rank parameterizations can be compared in a controlled environment. Second, the study is extended to reinforcement learning using stable-baselines-style implementations, including experiments with artificially increased input and output dimensions to assess whether low-rank policies remain effective for more general dynamics and policy classes. The empirical focus is on robustness with respect to parametric uncertainty, multiplicative control noise, measurement noise, and external disturbances. Optional theoretical analysis may complement the experiments, but the main goal is a systematic numerical comparison of full-rank and low-rank policies.

Tasks:

1. Familiarization with the linear quadratic regulator (LQR) problem, reinforcement learning, policy gradients, and the literature on structured and low-rank feedback control [7, 1, 8]
2. Implementation of full-rank and low-rank feedback parameterizations for the LQR problem, followed by an experimental comparison of fixed, learned, and model-informed low-rank bases
3. Extension of the experiments to reinforcement learning using stable-baselines-style implementations and artificially increased input/output dimensions
4. Robustness evaluation with respect to parametric uncertainty, multiplicative control noise, measurement noise, and external disturbances
5. Discussion of the resulting policies with respect to performance, robustness, and the trade-off between expressiveness and parameter efficiency

References

[1] Y. Park, R. A. Rossi, Z. Wen, G. Wu, and H. Zhao, "Structured policy iteration for linear quadratic regulator," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 7521–7531. [Online]. Available: https://proceedings.mlr.press/v119/park20a.html

[2] F. Zhao, F. Dörfler, and K. You, "Data-enabled policy optimization for the linear quadratic regulator," 2023. [Online]. Available: https://arxiv.org/abs/2303.17958

[3] S. Talebi, Y. Zheng, S. Kraisler, N. Li, and M. Mesbahi, "Policy optimization in control: Geometry and algorithmic implications," 2024. [Online]. Available: https://arxiv.org/abs/2406.04243

[4] N. Ay, "Geometric design principles for brains of embodied agents," *Künstliche Intelligenz*, vol. 29, no. 4, pp. 389–399, 2015. [Online]. Available: https://doi.org/10.1007/s13218-015-0382-z

[5] G. Montúfar, K. Ghazi-Zahedi, and N. Ay, "A theory of cheap control in embodied systems," *PLOS Computational Biology*, vol. 11, no. 9, p. e1004427, 2015. [Online]. Available: https://doi.org/10.1371/journal.pcbi.1004427

[6] ——, "Geometry and determinism of optimal stationary control in partially observable markov decision processes," 2015. [Online]. Available: https://arxiv.org/abs/1503.07206

[7] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*. Dover Publications, 1990.

[8] S. Takakura and K. Sato, "Structured output feedback control for linear quadratic regulator using policy gradient method," 2022. [Online]. Available: https://arxiv.org/abs/2203.00289

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